

Simulation of the Continuous Curvilinear Capsulorhexis Procedure

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Abstract. The capsulorhexis procedure is a key part of cataract surgery. During capsulorhexis, a circular opening is torn into the lens capsule. For managing a circular tear, the medical literature details a set of instructions for the inexperienced surgeon. We have translated these instructions into a descriptive algorithm that simulates the behavior of the capsulorhexis tear. The lens capsule is modeled by a mass-spring mesh. To enable smooth tear propagation within the mesh, we adapted an algorithm originally developed for interactive cutting in triangulated surfaces that constantly re-triangulates around the tear end. The capsulorhexis procedure is implemented as a module for the ophthalmosurgical training simulator EYESI.¹

1 Introduction

Pathological or aging processes may cloud the lens of the human eye. Cataract surgery replaces the clouded lens (the *cataract*) with an artificial lens implant. For that purpose, the *lens capsule* is opened, a clear, membrane-like structure that encloses the lens. After removing the lens, the open capsule serves as a container for the artificial lens. For preserving the stability of the capsule, Gimbel and Neuhaus developed the Continuous Curvilinear Capsulorhexis [4]: An initial radial tear is set by an incision instrument and defines radius and position of the opening. The loosened membrane patch is grabbed by a forceps or a hook and is moved in a way that the tear proceeds circularly (see Fig. 1). The anatomy of the eye makes this task difficult: The equator of the human lens is adnate with *zonular fibers* that are under constant tension and control the focusing of the eye. This tension also affects the lens capsule so that the capsulorhexis tear tends to propagate towards the periphery instead of following the direction of pulling.

To teach an inexperienced surgeon how to control the tear, we developed a capsulorhexis module for the ophthalmosurgical training simulator EYESI [14]

¹ EYESI is a product of VRmagic GmbH, Mannheim, Germany.

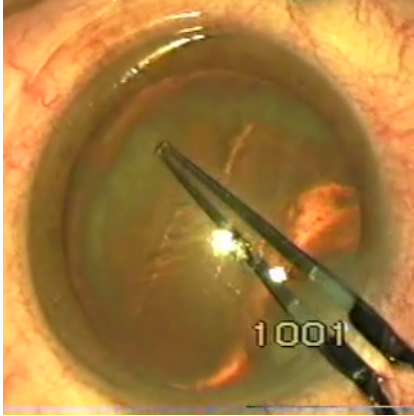


Fig. 1. A real capsulorhexis

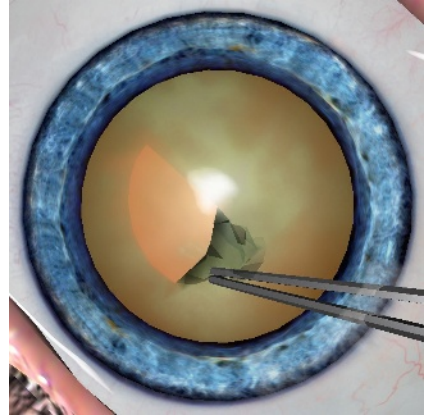


Fig. 2. A virtual capsulorhexis

(see Fig. 2). A mass-spring mesh serves as model for the deformation of the lens capsule. An algorithm for descriptive tear propagation in combination with an algorithm for interactive cutting models the capsulorhexis tear.

After a survey of related work, we present our implementation in Sect. 3 that cuts into two subsections: Sect. 3.1 concerns the descriptive algorithm, Sect. 3.2 describes the mesh adjustments which are necessary for a smooth tear propagation. Presentation of the results and an outlook conclude the paper.

2 Related Work

Related work covers developments of cataract surgery simulators within other research groups, modeling techniques concerning the simulation of deformable objects, and algorithms for cutting in triangulated surfaces.

2.1 Cataract Surgery Simulators

Several current research projects aim at realizing a cataract surgery simulator. The capsulorhexis procedure *“is universally acknowledged as one of the fundamental elements of modern phaco surgical techniques”* [12]. Hence, all these projects include the development of a capsulorhexis training module. However, the available publications just outline the implemented features and do not explicitly discuss the modeling of the capsulorhexis tear.

The Center for Advanced Studies, Research and Development in Sardinia is working on the EYESIM project. A mass-spring mesh is used for the simulation of the lens capsule, tearing is modeled by breaking overstretched springs [1]. The INRIA research project-team ALCOVE is developing a cataract surgery simulator that provides a capsulorhexis training based as well on a tearable mass-spring mesh [11]. This approach is also applied within the capsulorhexis

module of Webster et al. [15]. Recently, Webster et al. ported their module to the open API of the EYESI simulator which also served as platform for our implementation.

2.2 Physically Based Deformable Models

For the simulation of deformable materials, a number of physically based models have been established [8]. Recently, energy minimization models and meshless simulations have gained importance [13,7]. Mass-spring and Finite Element Models (FEM) are the most common methods used in surgical simulation [5]. The main advantage of FEM is that they allow physically accurate simulations based on the laws of continuum mechanics. To accelerate FEM for real-time applications, it is custom to linearize the involved strain and stress tensor. Especially concerning the simulation of highly deformable objects, this simplification may lead to physically incorrect and visually disturbing effects.

2.3 Cutting in Triangulated Surfaces

The smooth and steady propagation of a single tear requires a continuous re-triangulation around the tear end. In the literature, this problem is usually associated with cutting approaches. In this context, subdivision methods are very popular which refine those elements that are in contact with a scalpel [2,6]. These methods typically increase the mesh size and tend to produce degenerated elements. Ganovelli and O’Sullivan reduced the number of degenerated triangles by collapsing very short edges [3]. Since this solution does not repair all inconsistencies, Nienhuys and van der Stappen proposed to combine local mesh refinement with local on-the-fly Delaunay Triangulation [10]. It enables smooth extension of the cutting edge while maintaining the mesh quality. Therefore, we use this algorithm to handle topological changes along the capsulorhexis tear (see Sect. 3.2).

3 Simulating the Capsulorhexis Procedure

During capsulorhexis, the detached membrane patch is folded and crumpled by the surgical instrument. The stress distribution within the lens capsule determines the propagation of the tear. A physically accurate simulation of the lens capsule could facilitate the physically accurate prediction of the tear propagation. This would require a non-linear model that, in addition, must deal with topological changes along the tear. However, the computational cost of such a model is too high for the necessary update rate of a real-time application ($\geq 30\text{Hz}$). Thus, we subdivide the problem and separate the simulation of the tear from the simulation of the lens capsule.

Based on the medical literature, we have developed a descriptive algorithm that controls the propagation of the capsulorhexis tear (see Sect. 3.1). Thus, the

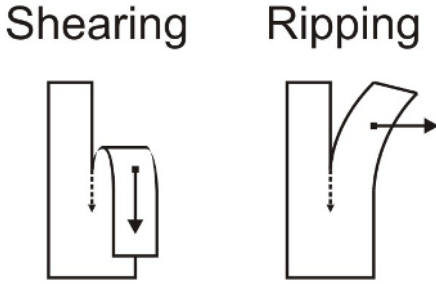


Fig. 3. Capsulorhexis techniques

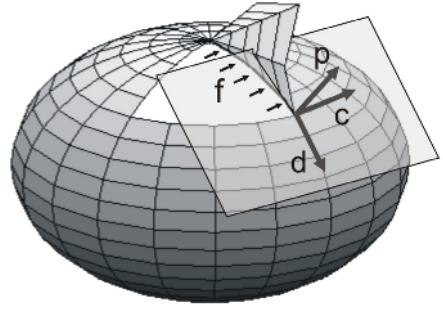


Fig. 4. Tangent plane on the lens

simulation of the membrane patch is of secondary importance so that a mass-spring mesh is quite adequate as basic model for the lens capsule. Stress within the membrane patch is only considered to trigger the tearing process. Apart from that, direction and velocity of the tear propagation are completely independent from the simulation of the lens capsule.

Each time a new tear direction has been calculated, the tear end propagates a small constant distance along this direction. Subsequently, the mass-spring mesh is updated accordingly (see Sect. 3.2).

3.1 Descriptive Propagation of the Capsulorhexis Tear

As the lens capsule is under constant tension, the capsulorhexis tear tends to propagate towards the periphery of the lens. To counter this drift and to manage a circular tearing edge, the technical literature describes two different techniques: *shearing* and *ripping*. Seibel [12] characterizes these two techniques as follows: In order to trigger shearing (Fig. 3, left), the surgeon has to flatten the membrane patch such that the inner surface of the membrane points upwards. Pulling the folded patch causes the capsulorhexis tear to follow approximately the movement of the instrument. However, shearing only enables small-sized deviations from the current tear direction and is unsuitable for the rescue of an already misrouted tear. In contrast, ripping (Fig. 3, right) enables relatively abrupt changes in the tear propagation. In order to switch to ripping, the surgeon has to unfold the patch such that its inner surface faces the lens. To achieve a circular tear propagation during ripping, the surgeon must pull towards the center of the lens. This behavior appears strange and the inexperienced surgeon needs practice to get used to it.

In our implementation, a parameter called **Indicator** determines whether the surgeon initiates shearing or ripping. We observe those triangles of the mass-spring mesh that are both adjacent to the tear end and part of the already detached membrane patch. If the according triangle normals face the lens surface, shearing will be triggered (**Indicator** := **Shearing**) otherwise ripping (**Indicator** := **Ripping**).

Basic Parameters of Tear Propagation. Concerning both shearing and ripping, the new direction for the propagating tear depends on the current tear direction and the direction of pulling. These parameters have to be measured in an appropriate manner: Since the lens capsule is attached to the lens with its smooth, convex surface, the current end of the tear and the according surface normal of the lens define a tangent plane which we use as subspace for our calculations. Projection of the tearing edge onto the plane yields the current tear direction **CurrDir** (vector **c** in Fig. 4) and projection of the instrument’s tip yields the current direction of pulling **PullDir** (vector **p** in Fig. 4). Also the direction of the drift **DriftDir**, that points towards the periphery, can be defined within the tangent plane (vector **d** in Fig. 4). The new direction of the tear propagation is determined by rotating **CurrDir** around the surface normal of the tangent plane. The final rotation angle is the sum of two signed angles: **DriftAngle** denotes a rotation towards **DriftDir**, **PullAngle** denotes a rotation towards **PullDir**. The calculation of these two rotation angles depends on the value of **Indicator**.

Shearing. If the surgeon applies this technique, the drift only has a minor effect on the tear propagation. Thus, during shearing **DriftAngle** is set to a small, constant value for the rotation of **CurrDir** towards **DriftDir**. On the other hand, the value of **PullAngle** corresponds to the angle enclosed by **CurrDir** and **PullDir** and defines the rotation of **CurrDir** towards **PullDir**. Since shearing may provoke only small changes in **CurrDir**, **PullAngle** must not exceed a predefined maximum. The sum of both **DriftAngle** and **PullAngle** yields the final rotation angle for **CurrDir**.

Ripping. During ripping, the drift strongly affects the propagating tear. Therefore, **DriftAngle** is higher than during shearing. Experienced surgeons describe that ripping sometimes requires a direction of pulling that is nearly perpendicular to the desired tear propagation. In order to model this behavior, **PullAngle** is only assigned a predefined fractional amount of the angle enclosed by **CurrDir** and **PullDir**. Since ripping enables relatively abrupt turnarounds of **CurrDir**, **PullAngle** is constrained by a maximum that is higher than the according maximum during shearing.

3.2 Mesh Adjustment During Tearing

Every time the position of the tear end node is changed, the surrounding triangles have to be adjusted. In addition, another piece of the lens capsule will be detached from the lens and the membrane patch grows by this piece.

Progress of the Tearing Edge. The cutting algorithm proposed by Nienhuys et al. defines a scalpel that can be moved by the user [10]. In our training module it is the tearing algorithm that controls the movement of the “scalpel”. Aside from this point, the original cutting algorithm can be used for our purpose, thus we describe the remeshing process only briefly:²

² We refer the reader to the technical report [9] that is more detailed than the homonymous paper [10].

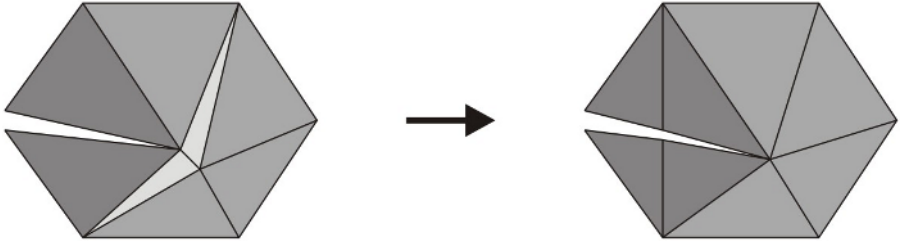


Fig. 5. Nodes close to the tear end will be removed and long tearing edges will be split

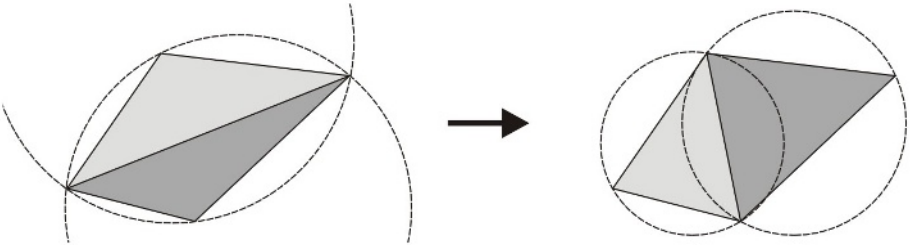


Fig. 6. An edge that is not Delaunay will be flipped

1. If the distance between the propagating tear end node and an adjacent node is less than a predefined threshold parameter, the adjacent node will be removed. Thus, the two light-gray triangles in Fig. 5 disappear.
2. If the length of the tearing edge behind the tear end is greater than a predefined threshold parameter, the corresponding edges will be split. Thus, the algorithm subdivides the two dark-gray triangles in Fig. 5.
3. To assure mesh quality around the tear end node, the *Delaunay Criterion* is used. An edge is called *Delaunay* if the circumcircles of the two adjacent triangles do not contain the respective remaining triangle node. If an edge is not Delaunay, it will be *flipped* as shown in Fig. 6 so that the minimal angle within the two triangles increases. Every time the tear end node progresses, the surrounding edges will be checked and flipped if necessary.

Progress of the Membrane Patch. The membrane patch defines a *tearing front* that separates the detached membrane from the part of the lens capsule which is still attached to the lens surface. The tearing front is illustrated in Fig. 4 by **f** and has to propagate along with the tear end node which always defines the outer end of the front. As long as the capsulorhexis covers less than 360 degrees, the inner end of the tearing front is given by a node in the middle of the mesh. Fig. 7 exemplifies this state of a virtual capsulorhexis and shows a wire frame of the lens capsule. If the diameter of the capsulorhexis is too small, the surgeon must increase it by continuing the capsulorhexis beyond 360 degrees.

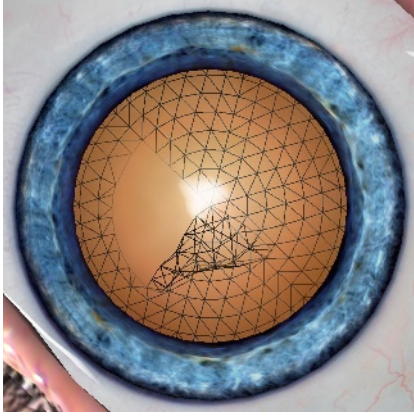


Fig. 7. A capsulorhexis that covers less than 360 degrees (number of triangles: 486)

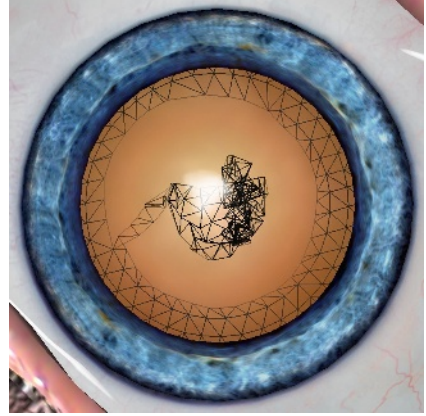


Fig. 8. Continuation of the capsulorhexis shown in Fig. 7 beyond 360 degrees (number of triangles: 530)

Once the capsulorhexis exceeds 360 degrees, the inner node of the tearing front also detaches from the lens as indicated in Fig. 8.

Every node of the mesh holds a boolean flag **Attached**. If **Attached** is **TRUE**, the corresponding node is excluded from the mass-spring simulation and thus, attached to its initial position on the lens surface. A flag switches from **TRUE** to **FALSE** at the time when the tearing front passes the corresponding node. The tearing front is given by a line segment that connects the inner and outer end of the front and that is projected onto the lens surface.

4 Results

The proposed tearing algorithm models the specific behavior of a capsulorhexis tear, including the distinction between shearing and ripping: If the surgeon proceeds inattentively, an unintentional switch from shearing to ripping may occur. At this moment, the direction of pulling is not adequate for ripping which leads to a tear propagation towards the periphery – an effect that is described in the literature. On the other hand, the medical literature recommends controlled ripping to redirect a misrouted capsulorhexis tear. Within our model, only ripping allows for relatively abrupt turnarounds of the current tear direction. Thus, the surgeon learns to apply planned ripping maneuvers for the rescue of an already errant capsulorhexis tear.

The combination of the algorithm with online remeshing around the tear end node enables the simulation of a steady tear propagation that hides the polygonal structure of the mass-spring mesh. However, the resolution of the mesh must be

adapted to the threshold parameters of the underlying cutting algorithm. Fig. 7 and Fig. 8 exemplify a resulting tearing edge.

Due to the computational efficiency of the mass-spring model and the descriptive algorithm, the complete EYESI module fulfills the real-time condition. As the propagation of the tear is separated from the simulation of the lens capsule, the choice of the mass-spring parameters and the time integration scheme is relatively uncritical.

5 Discussion and Outlook

We have combined an algorithm for descriptive tear propagation with an algorithm for interactive cutting in order to simulate a tearing membrane. The implementation runs as a cataract training module on the EYESI system and is already in use for surgical training.

The validation by ophthalmosurgeons has shown that the simulated behavior of the tear is realistic enough to meet the demands on a capsulorhexis training. However, the descriptive nature of the model still restricts the possibilities of the surgeon: The descriptive algorithm is mapped onto a single tear that is automatically initialized when starting the module. Currently, it is not possible to incise the membrane during the simulated procedure at arbitrary positions.

In order to enable the interactive use of an incision instrument, current work includes the simulation of multiple tears. This will allow for the integration of the initial capsulorhexis incision as part of the training task. Besides, we are working on the simulation of viscoelastic fluid injection that affects the behavior of the tissue.

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